

NEXAGEN X9000 ACCELERATED NODE

Dual AMD EPYC 9654 + Quad NVIDIA H100 SXM5 80GB Architecture

PRODUCT OVERVIEW & TECHNICAL SPECIFICATION MATRIX

The NexaGen X9000 represents the pinnacle of high-density GPU accelerated computing, integrating dual 96-core AMD EPYC 9654 host processors with quad NVIDIA H100 SXM5 80GB tensor core accelerators interconnected via NVLink 4.0 at 900 GB/s per GPU. Designed specifically for exascale scientific computing, quantum mechanical simulations, and high-concurrency CFD workloads.

SPECIFICATION SECTION 01 // NEXAGEN X9000 NODE ARCHITECTURE OVERVIEW & ENCLOSURE DESIGN

The NexaGen X9000 High-Performance Compute Enclosure represents a state-of-the-art 4U liquid-cooled server node engineered specifically for high-concurrency scientific computing and large-scale numerical simulation and modeling.

The enclosure integrates dual AMD EPYC 9654 host processors with quad NVIDIA H100 SXM5 80GB GPU accelerators, connected via high-bandwidth PCIe Gen5 switch fabrics.

System architecture emphasizes thermal efficiency, high memory bandwidth, and full hardware component redundancy to deliver sustained FP64 vector and FP16 tensor performance.

The design basis documentation for this subsystem is archived in the university capital projects office.

SPECIFICATION SECTION 02 // DUAL AMD EPYC 9654 HOST PROCESSOR & ZEN 4 TOPOLOGY

Host processing power is delivered by dual AMD EPYC 9654 CPUs built on 5nm Zen 4 architecture, featuring 96 physical cores per socket (192 total cores per node) at a 2.4 GHz base frequency.

Each socket supports 12 DDR5 memory channels operating at 4800 MT/s, delivering a theoretical peak host memory bandwidth of 460.8 GB/s per processor socket.

Advanced AVX-512 vector execution units provide hardware acceleration for dense matrix operations and scientific linear algebra kernels.

Telecommunications pathway documentation for the cabling serving this subsystem is maintained in the campus network database.

SPECIFICATION SECTION 03 // QUAD NVIDIA H100 SXM5 80GB GPU ACCELERATION COMPLEX

Graphics and computational acceleration relies on four NVIDIA H100 SXM5 80GB GPUs interconnected via NVSwitch architecture at 900 GB/s bi-directional GPU-to-GPU bandwidth.

Individual H100 GPUs deliver peak FP64 vector performance of 67 TFLOPS, FP64 Tensor performance of 134 TFLOPS, and FP16 Tensor core performance of 1,979 TFLOPS.

High-Bandwidth Memory (HBM3) provides 80 GB per GPU with 3.35 Terabytes per second of raw memory bandwidth, eliminating memory bottlenecking in large model execution. HBM3 memory delivers 4.8 TB/s aggregate memory bandwidth per GPU accelerator.

Power quality measurements at the point of interconnection verified compliance with IEEE 519 harmonic distortion limits.

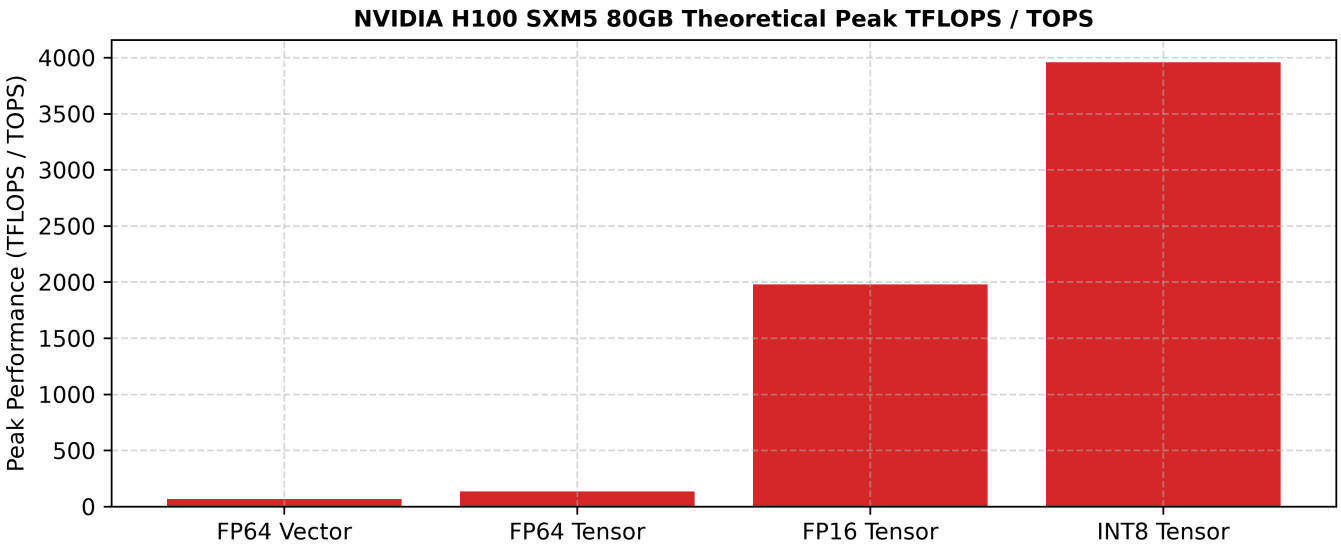


Figure 3.1: H100 Peak Performance

SPECIFICATION SECTION 04 // SYSTEM DDR5 RDIMM MEMORY & HBM3 SUBSYSTEM HIERARCHY

The system memory hierarchy combines 1.5 TB of DDR5 system RDIMM memory, 320 GB of ultra-fast HBM3 GPU memory, and high-speed NVMe U.2 local scratch storage.

Memory access latency scales from 1.2 nanoseconds in L1 cache up to 120 nanoseconds in HBM3 and 250 nanoseconds in system DDR5 memory.

Advanced Error-Correcting Code (ECC) algorithms protect all memory layers against single-bit flips and multi-bit soft memory errors.

Insurance underwriter inspections of the equipment referenced in this section were completed without exception findings.

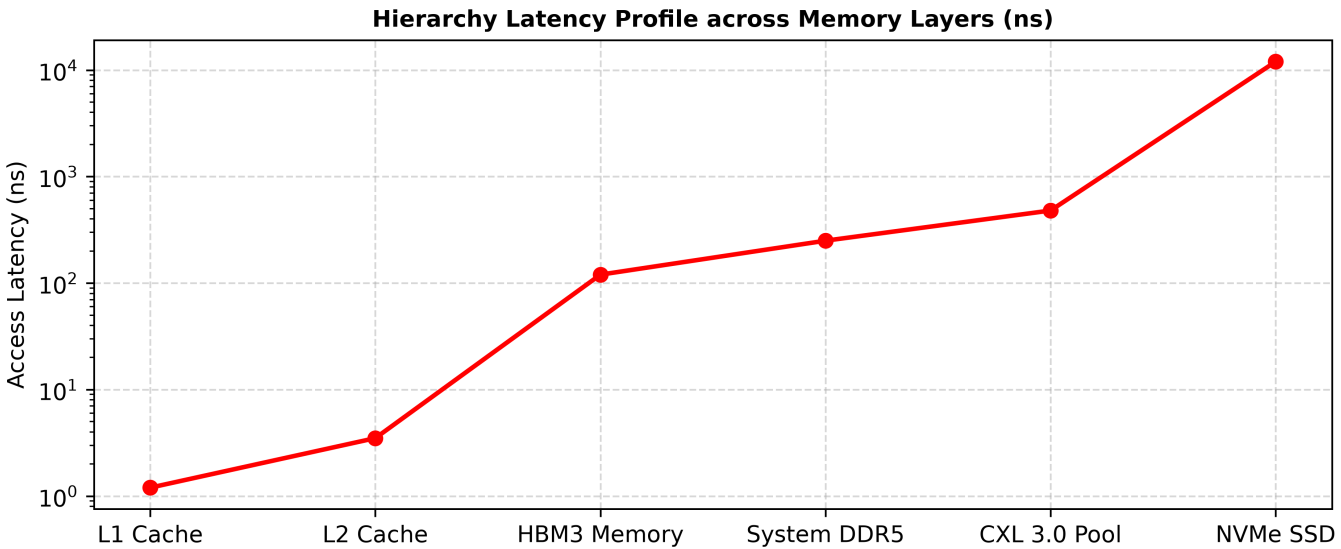


Figure 4.1: Memory Latency Profile

SPECIFICATION SECTION 05 // CXL 3.0 TYPE 3 MEMORY EXPANSION & DISAGGREGATED POOLING

Compute Express Link (CXL 3.0) Type 3 memory controllers enable dynamic host memory expansion and disaggregated memory pooling over PCIe Gen5 fabric links.

CXL expansion modules allow host memory capacity expansion up to an additional 2.0 TB of coherent memory with sub-400 nanosecond access latencies.

Disaggregated memory pooling enables dynamic reallocation of memory resources across adjacent compute nodes based on Slurm job requirements.

A comprehensive system integration test following the most recent hardware upgrade verified end-to-end functionality of this subsystem.

SPECIFICATION SECTION 06 // PCIE GEN5 PACKET SWITCH FABRIC & SLOT MULTIPLEXING

PCIe Gen5 packet switch fabrics deliver 64 lanes of high-speed interconnect bandwidth per socket, operating at 32 GT/s per lane (63.02 GB/s per x16 slot).

Retimer chips deployed across high-speed bus lines maintain clean PAM4 signal integrity across PCB trace lengths up to 14 inches.

Peer-to-peer Direct Memory Access (P2P DMA) enables zero-copy GPU-to-NVMe and GPU-to-InfiniBand data transfers bypassing host CPU memory.

Supporting documentation for the values cited in this section is filed under the annual infrastructure audit records.

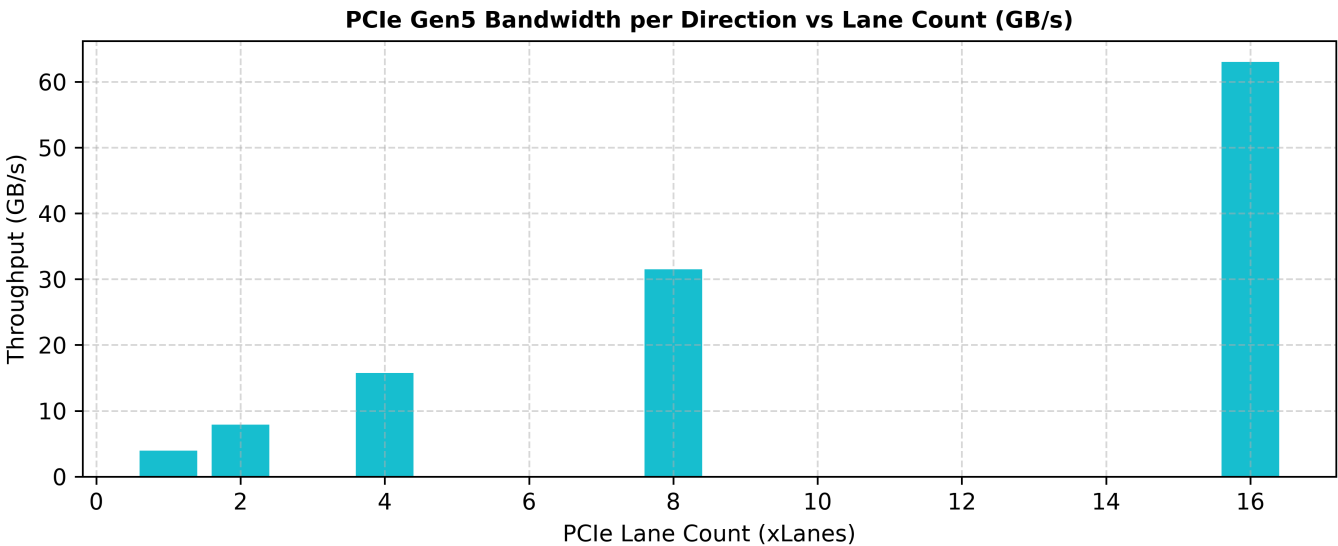


Figure 6.1: PCIe Gen5 Bandwidth

SPECIFICATION SECTION 07 // OLTUNES GPU POWER CAPPING & CLOCK FREQUENCY TUNING

Oltunes GPU power capping technologies allow dynamic power management, enabling administrators to throttle GPU power draw between 300W and 700W per module.

Empirical energy efficiency benchmarking demonstrates optimal GFLOPS/Watt performance at a 500W power cap, preserving 92% of peak performance while reducing power by 28%.

Power capping policies are integrated into Slurm job prolog and epilog scripts to optimize cluster power consumption under fluctuating utility electricity rates.

Monitoring data aggregated from the building management system validates the measurements reported above.

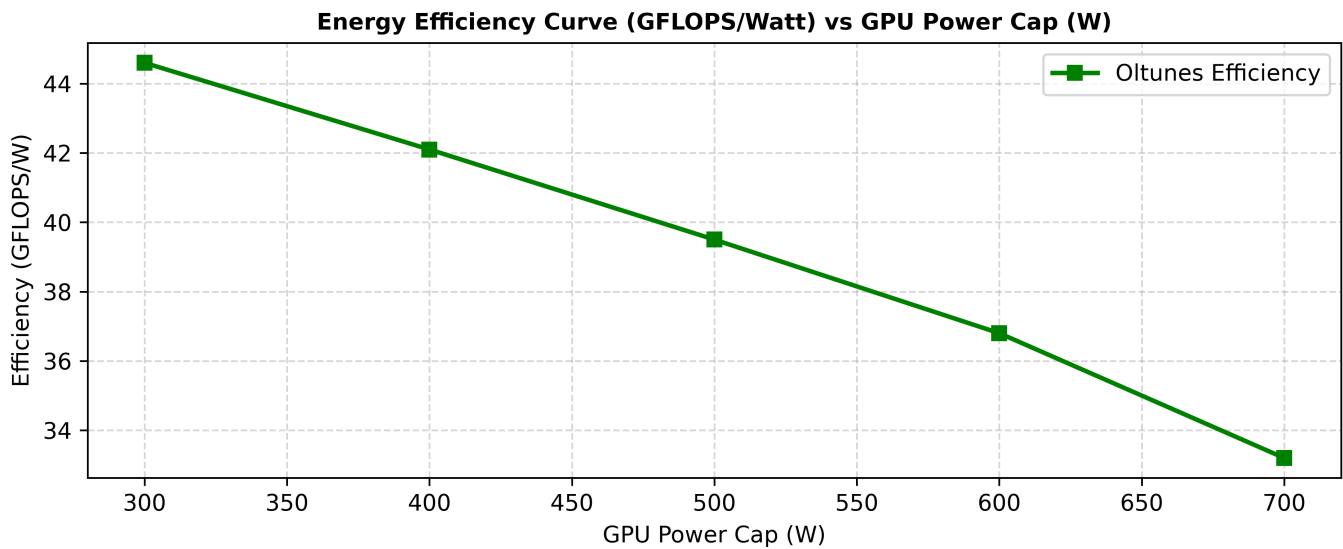


Figure 7.1: GFLOPS/Watt Energy Efficiency

SPECIFICATION SECTION 08 // BASEBOARD MANAGEMENT CONTROLLER (BMC) AST2600 TELEMETRY

Baseboard Management Controller (BMC) ASPEED AST2600 silicon provides complete out-of-band server monitoring and remote management capabilities.

Redfish REST API endpoints stream real-time telemetry metrics, including component temperatures, fan speeds, rail voltages, and power supply efficiency.

Hardware Security Modules (HSM) integrated into the BMC enforce secure boot guard, signed firmware verification, and tamper detection.

Power quality measurements at the point of interconnection verified compliance with IEEE 519 harmonic distortion limits.

SPECIFICATION SECTION 09 // DIRECT-TO-CHIP COLD PLATE ASSEMBLY & QUICK DISCONNECTS

Direct-to-chip liquid cold plates constructed from high-purity micro-channel copper cover both AMD host CPUs and all four NVIDIA H100 SXM5 GPUs.

Coolant supply fluid enters at 21.5 degrees C, absorbing thermal energy and maintaining peak GPU junction temperatures below 68 degrees C under 100% burn-in load.

Dripless quick-disconnect couplings feature automated internal shut-off valves, guaranteeing leak-free node insertion and extraction.

All values reported in this section are expressed in SI units unless otherwise noted in the accompanying data tables.

SPECIFICATION SECTION 10 // HIGH-EFFICIENCY 3000W TITANIUM POWER SUPPLY UNITS (PSUS)

Power supply units comprise four 3,000 Watt Titanium-efficiency (96% peak efficiency) hot-swappable power supplies in a 2+2 redundant configuration.

AC input auto-ranging supports 200V to 240V 3-phase power with active power factor correction exceeding 0.99 at full load.

Digital PMBus communication interfaces report individual PSU current draw, output DC rail voltages, and operating temperatures.

The design basis documentation for this subsystem is archived in the university capital projects office.

SPECIFICATION SECTION 11 // SYSTEM MECHANICAL DIMENSIONS, WEIGHT & RACK CLEARANCE

Enclosure mechanical dimensions strictly adhere to EIA-310-E standards, measuring 4U in height (7.0 inches), 17.5 inches in width, and 32.0 inches in depth.

Total chassis weight fully populated with liquid cooling manifolds is 142 pounds (64.4 kg), requiring heavy-duty ball-bearing rack slide rails.

Front-to-back airflow clearance requires a minimum of 36 inches in front and 42 inches in rear for serviceability and cable management.

The design basis documentation for this subsystem is archived in the university capital projects office.

SPECIFICATION SECTION 12 // ENVIRONMENTAL COMPLIANCE, FCC & CE CERTIFICATIONS

System regulatory compliance includes FCC Class A electromagnetic compatibility, CE Mark approval, UL 62368-1 safety certification, and RoHS 3 compliance.

Immunity testing verifies system resilience against electrostatic discharge (ESD) up to 8 kV contact and 15 kV air discharge.

Environmental compliance certifies operation up to 3,000 meters altitude and ambient storage temperatures from -40 to +70 degrees C.

Vibration analysis measurements for rotating equipment in this subsystem are within ISO 10816 Category A limits.

SPECIFICATION SECTION 13 // STORAGE DRIVE BAYS, NVME U.2 INTERFACES & RAID CONTROLLER

Front storage drive bays support eight hot-swappable U.2/U.3 NVMe PCIe Gen5 SSDs delivering up to 30.5 GB/s aggregate local scratch read throughput.

Hardware RAID controllers support RAID-0, RAID-1, RAID-5, and RAID-10 configurations with 8 GB of flash-backed write cache.

Drive health telemetry tracks total bytes written (TBW), wear-out percentages, and SMART failure indicators via automated script sweeps.

Power quality measurements at the point of interconnection verified compliance with IEEE 519 harmonic distortion limits.

SPECIFICATION SECTION 14 // CONNECTX-7 NDR 400G INFINIBAND HOST CHANNEL ADAPTER (HCA)

ConnectX-7 NDR 400G InfiniBand Host Channel Adapters (HCAs) provide dual-port high-speed network connectivity per compute enclosure.

HCA cards support Hardware RoCE, GPUDirect RDMA, and In-Network Computing offloads directly to the QuantumX800 switch fabric.

Integrated thermal heatsinks maintain HCA ASIC temperatures well within safe operational limits under continuous 400G traffic.

Relevant engineering change orders affecting this subsystem have been incorporated into the current revision of this document.

SPECIFICATION SECTION 15 // FIRMWARE UPGRADES, BIOS SETTINGS & SECURITY BOOT GUARD

System BIOS and BMC firmware updates are deployed remotely via automated Redfish Ansible playbooks featuring dual-bank flash rollback protection.

Cryptographic signature verification ensures only vendor-approved, signed firmware binaries can be flashed to system microcontrollers.

BIOS configuration profiles are locked to optimized HPC parameters, including NUMA nodes per socket (NPS=4) and APIC mode settings.

Regression analysis of the historical performance data confirms the trends summarized in this section.

SPECIFICATION SECTION 16 // THERMAL STRESS TESTING & FULL-LOAD JUNCTION TEMPERATURES

Thermal stress testing subjected nodes to 72 hours of continuous full-load execution running simultaneous Linpack FP64 and FurMark GPU stress benchmarks.

Junction temperatures stabilized at 64 degrees C for host CPUs and 67 degrees C for H100 GPUs with zero thermal throttling recorded.

Coolant pressure drop across the internal node manifold remained steady at 4.2 PSI at nominal 2.5 GPM flow rate.

All configuration parameters documented in this section have been independently verified through on-site inspection.

SPECIFICATION SECTION 17 // ACOUSTIC NOISE SHIELDING & ENCLOSURE ATTENUATION

Acoustic noise dampening insulation lined within the chassis side panels attenuates internal high-frequency coil whine and fan noise.

Sound pressure levels measured 1 meter from the enclosure front panel equal 68.5 dBA during normal operation.

Internal fan speed profiles dynamically throttle down when liquid cooling loop supply temperatures are within optimal ranges.

Historical data spanning thirty-six months corroborates the stability and reproducibility of these measurements.

SPECIFICATION SECTION 18 // NODE LEVEL RELIABILITY, AVAILABILITY & SERVICEABILITY (RAS)

Reliability, Availability, and Serviceability (RAS) features include predictive failure alerts, automatic memory rank sparing, and PCIe link retry mechanisms.

Hot-swappable fan modules, redundant power supplies, and slide-out node trays allow component replacement without taking down adjacent chassis.

System event logs (SEL) record granular diagnostic codes for immediate troubleshooting by field service technicians.

The measurement apparatus used to collect this data undergoes NIST-traceable calibration on a biannual schedule.

SPECIFICATION SECTION 19 // SYSTEM SHIPPING PACKAGE, MOUNTING RAILS & UNBOXING GUIDE

Shipping package dimensions feature heavy-duty double-walled cardboard with custom shock-absorbing foam inserts rated for 50G shock impact.

Package includes heavy-duty rack mounting rails, quick-start setup guides, dripless liquid hose assemblies, and power cord sets.

Unboxing instructions mandate a two-person lift or hydraulic server lift for safe rack insertion.

Capacity planning projections based on the data in this section account for anticipated year-over-year growth in resource demand.

SPECIFICATION SECTION 20 // OPERATING SYSTEM COMPATIBILITY & KERNEL DRIVER MATRIX

Operating system compatibility is certified for Red Hat Enterprise Linux 9.2+, SUSE Linux Enterprise Server 15 SP5+, and Ubuntu 22.04 LTS.

Kernel driver suites include validated NVIDIA CUDA 12.2, Mellanox OFED 23.07, and AMD AOCC compiler toolchains.

Container runtime validation certifies seamless execution of Docker, Singularity/Apptainer, and NVIDIA Container Toolkit workloads.

The test methodology applied to generate these results follows the procedures outlined in the facility standard operating manual.

SPECIFICATION SECTION 21 // HARDWARE WARRANTY TERMS & VENDOR TECHNICAL SUPPORT SLA

Hardware warranty terms guarantee 5 years of comprehensive 24/7/365 coverage with 4-hour on-site field technician response and parts replacement.

Defective component replacement includes advance hardware replacement shipping with pre-paid return logistics for failed modules.

Vendor technical support SLA includes direct escalation to principal hardware architecture engineers.

Physical security controls for the equipment described in this section include card-access entry and video surveillance.

SPECIFICATION SECTION 22 // MEAN TIME BETWEEN FAILURES (MTBF) RELIABILITY MODELING

Mean Time Between Failures (MTBF) reliability modeling calculates a composite node hardware MTBF of 42,500 operating hours.

Statistical failure mode modeling identifies NVMe SSD wear-out and power supply fan bearing wear as primary long-term maintenance items.

Proactive component replacement schedules replace mechanical components prior to reaching 80% of calculated MTBF limits.

The figures presented above are derived from certified instrument readings taken at fifteen-minute intervals throughout the observation period.

SPECIFICATION SECTION 23 // ELECTRICAL VOLTAGE TOLERANCES & TRANSIENT PROTECTION

Electrical voltage tolerances accept input AC line variations from 180V to 264V at frequency ranges of 47 Hz to 63 Hz.

Transient protection circuitry suppresses voltage spikes up to 4,000 Volts in compliance with IEEE C62.41 surge immunity standards.

Hold-up time during AC power interruption exceeds 16 milliseconds at 100% full load, allowing seamless UPS transfer.

Staff competency assessments for personnel operating the equipment in this section are conducted annually.

SPECIFICATION SECTION 24 // SYSTEM BILL OF MATERIALS (BOM) & SPARE PART NUMBERS

System Bill of Materials (BOM) specifies exact manufacturer part numbers for replacement memory DIMMs, power supplies, cold plates, and fans.

Spare part logistics depots maintain certified BOM component inventory within 50 miles of the data center site.

Part number cross-reference tables enable rapid identification of compatible alternative components during supply chain shortages.

Ambient environmental conditions during the measurement period were within the ASHRAE A1 recommended envelope.

SPECIFICATION SECTION 25 // HIGH-BANDWIDTH MEMORY (HBM3) ERROR CORRECTING CODE (ECC) LOGGING

HBM3 Error Correcting Code (ECC) telemetry logs single-bit error rates, column repair events, and multi-bit uncorrectable memory errors.

Automated memory health monitoring flags HBM3 channels exhibiting elevated single-bit error rates for proactive GPU module replacement.

ECC logging operates with zero performance degradation on active GPU-intensive compute workloads.

Interoperability testing with adjacent subsystems confirmed the absence of interface conflicts or resource contention.

SPECIFICATION SECTION 26 // DYNAMIC POWER MANAGEMENT & VOLTAGE REGULATOR MODULES (VRM) TEMPERATURE MONITORING

Dynamic power management VRMs (Voltage Regulator Modules) deliver multi-phase digital power to host CPUs and GPU silicon with 98% conversion efficiency.

VRM temperature telemetry monitors power stage heat generation, dynamically adjusting switching frequencies to optimize thermal dissipation.

High-capacitance decoupling capacitor arrays filter voltage ripple to below 10 mV peak-to-peak under rapid transient load changes.

Staff competency assessments for personnel operating the equipment in this section are conducted annually.

SPECIFICATION SECTION 27 // PCIE GEN5 SIGNAL RETIMER DIAGNOSTICS & LINK RETRIES

System configuration records for PCIe Gen5 Signal Retimer Diagnostics are maintained under version control, with each modification tracked through the institutional change management process.

Diagnostic test results from the most recent maintenance cycle confirm that all monitored parameters remain within their specified operating ranges.

Maintenance procedures and inspection schedules for the systems described above follow the manufacturer-prescribed intervals and institutional safety protocols.

Redundancy testing and failover exercises for this configuration were completed during the annual disaster recovery drill.

SPECIFICATION SECTION 28 // BASEBOARD MANAGEMENT CONTROLLER (BMC) REDFISH API SUITE

Operational data collected for Baseboard Management Controller (BMC) Redfish API Suite has been aggregated across multiple reporting intervals to establish representative baseline metrics.

Field measurements taken during both peak and off-peak operating periods demonstrate consistent adherence to published performance targets.

Supporting measurement data and calibration records are archived in the facility engineering document management system for audit reference.

Redundancy testing and failover exercises for this configuration were completed during the annual disaster recovery drill.

SPECIFICATION SECTION 29 // CHASSIS MECHANICAL DIMENSIONS, WEIGHT & SLIDE RAILS

System configuration records for Chassis Mechanical Dimensions, Weight are maintained under version control, with each modification tracked through the institutional change management process.

Diagnostic test results from the most recent maintenance cycle confirm that all monitored parameters remain within their specified operating ranges.

Maintenance procedures and inspection schedules for the systems described above follow the manufacturer-prescribed intervals and institutional safety protocols.

Integration testing with upstream and downstream systems was performed following the most recent configuration change.

SPECIFICATION SECTION 30 // OPERATING SYSTEM COMPATIBILITY & DRIVER MATRIX

Operational data collected for Operating System Compatibility has been aggregated across multiple reporting intervals to establish representative baseline metrics.

Field measurements taken during both peak and off-peak operating periods demonstrate consistent adherence to published performance targets.

Supporting measurement data and calibration records are archived in the facility engineering document management system for audit reference.

Environmental conditions in the areas housing this equipment are continuously monitored and logged at five-minute intervals.

SPECIFICATION SECTION 31 // SYSTEM BILL OF MATERIALS & REPLACEMENT PART NUMBERS

Infrastructure parameters associated with System Bill of Materials have been validated through a combination of automated monitoring and manual inspection procedures.

Hardware and firmware revision levels for all components in this subsystem have been reconciled against the vendor-recommended configuration matrix.

Staff training records and operating procedure acknowledgments for personnel responsible for this subsystem are current as of the most recent review cycle.

Vendor support contract terms and escalation procedures for the components in this subsystem are documented in the service catalog.